

PSimPy - Predictive and probabilistic simulation

Hu Zhao, Anil Yildiz, Nazanin Bagherinejad
Chair of Methods for Model-based Development in Computational Engineering
RWTH Aachen University



Summary

Robust GP emulator for your simulation

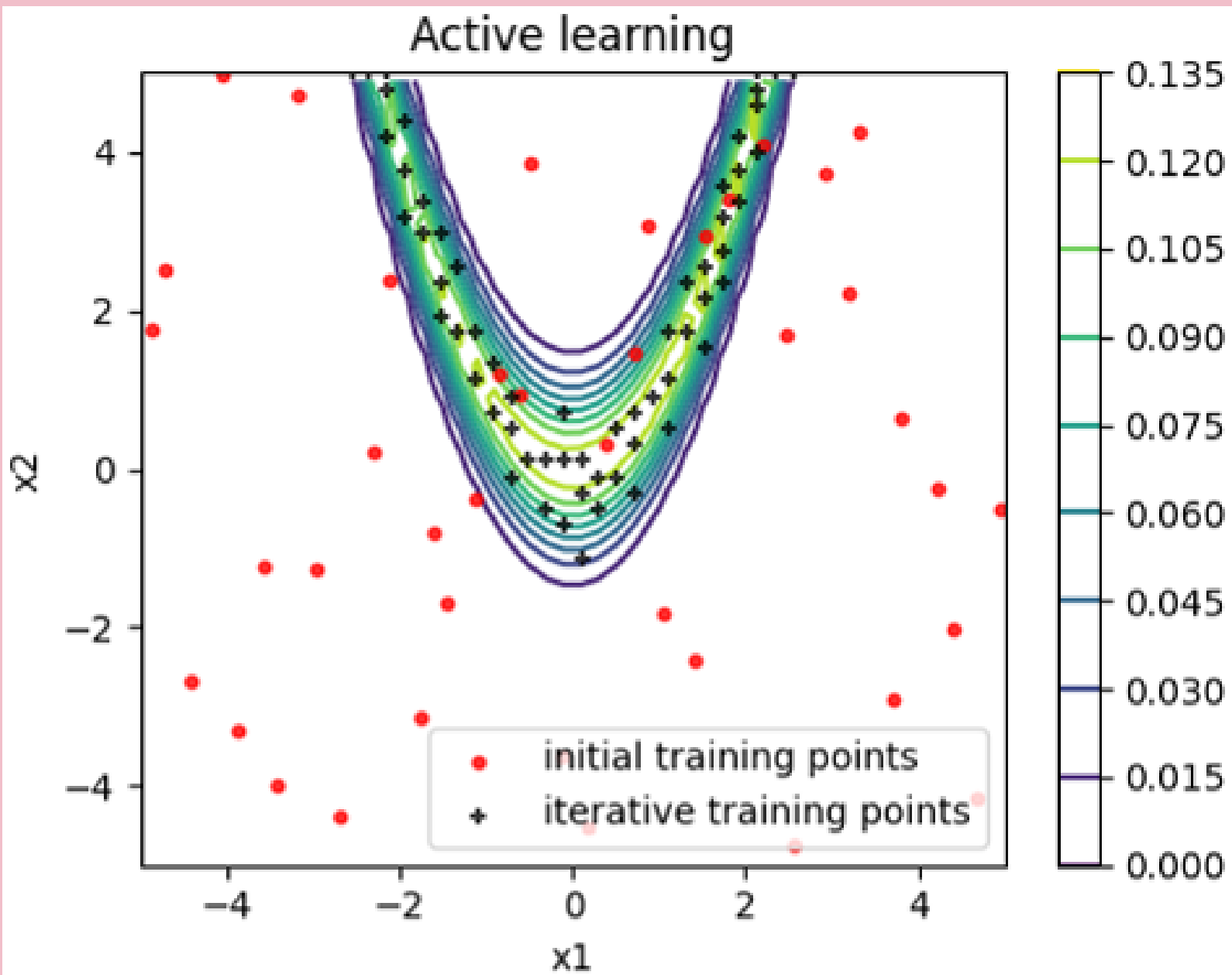
- Gaussian Process emulation
- Bayesian Calibration
- Global Sensitivity Analysis
- Efficient Sampling
- Active Learning

Lightweight Substitute = Fast Evaluations

	training	evaluation	2500 evals
simulator	-	32 min	56 days
emulator	4.5 days	0.05 sec	2 min

Software in Action

Quantify uncertainty. Identify key drivers.
Calibrate with data.



Want to build unified,
robust and sustainable
UQ workflows?

PSiPy^[1]



Reference

[1] H. Zhao, A. Yildiz, N. Bagherinejad, and J. Kowalski, “PSimPy: GP emulation-based sensitivity analysis, uncertainty quantification and calibration of landslide simulators,” in *14th International Conference on Applications of Statistics and Probability in Civil Engineering, ICASP14*, Dublin, Ireland, 2023.